PaddyGuard AI: A Multimodal AI-Based System for Rice Leaf Disease & Pest Detection, Treatment Recommendation and Knowledge Assistance

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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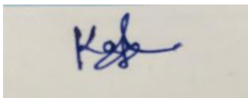
Keshan B K	IT22303820	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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ABSTRACT

Rice is one of the most important staple crops in many countries, yet rice farmers frequently experience significant yield losses due to diseases and pest infestations. Early identification of these issues and the selection of appropriate treatments are essential for maintaining crop health and productivity. However, many smallholder farmers have limited access to agricultural experts, diagnostic laboratories, and reliable information about safe pest and disease management practices. As a result, farmers often rely on delayed or inaccurate diagnosis, leading to ineffective treatments and excessive pesticide use that can negatively affect both crop yield and environmental sustainability.

Existing digital plant disease detection systems mainly rely on image-based analysis and often lack integrated decision support mechanisms that provide reliable treatment guidance. Furthermore, most available tools do not support multimodal inputs such as voice descriptions, which limits accessibility for farmers with low literacy levels. This highlights a significant research gap in the development of a comprehensive, farmer-centric system that combines intelligent diagnosis with actionable treatment advisory services.

To address these challenges, this research proposes **PaddyGuard AI**, a multimodal intelligent system for rice disease and pest management. The proposed solution integrates image-based deep learning models for disease and pest detection, voice-based symptom analysis using natural language processing, and a knowledge-driven treatment advisory system supported by an agricultural chatbot. The advisory system incorporates rule-based validation, a pesticide dosage calculator, and a retrieval-augmented generation (RAG) framework to deliver accurate and context-aware recommendations to farmers.

The system will be developed using technologies such as deep learning frameworks (PyTorch or TensorFlow), computer vision techniques, speech recognition models, natural language processing with transformer architectures, and a knowledge retrieval system for advisory responses. The expected outcome is a reliable and user-friendly platform capable of providing accurate diagnosis, safe treatment recommendations, and accessible voice-based guidance.

The proposed solution has the potential to improve crop productivity, reduce pesticide misuse, and support sustainable agricultural practices. Socially, it empowers farmers with accessible digital decision support, while commercially it creates opportunities for scalable agricultural advisory platforms and smart farming solutions.

Keywords: Rice Disease Detection, Agricultural Chatbot, Multimodal AI, Precision Agriculture, Treatment Advisory System

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
AI	Artificial Intelligence
CNN	Convolutional Neural Network
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
ASR	Automatic Speech Recognition
OOD	Out-of-Distribution

DSS	Decision Support System
IRRI	International Rice Research Institute
ML	Machine Learning
UI	User Interface

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1. INTRODUCTION

1.1 Background and Context

Rice is one of the most important staple crops in the world and plays a crucial role in ensuring global food security. Millions of farmers depend on rice cultivation as their primary source of livelihood. However, rice production is frequently threatened by various plant diseases and pest infestations that significantly reduce crop yield and quality. According to global agricultural reports, plant diseases and pests are responsible for nearly 20–40% of crop losses worldwide, creating serious economic and food security challenges. In many rice-producing Advisory Systems and Chatbot Component 20uggle to identify crop diseases and pests at an early stage and select appropriate treatment methods.

One of the major challenges faced by farmers is the limited access to agricultural expertise and reliable crop treatment guidance. Farmers often rely on personal experience, traditional knowledge, or informal advice when dealing with crop diseases. As a result, incorrect pesticide usage, delayed diagnosis, and inappropriate treatment practices frequently occur. These practices can increase production costs, cause environmental damage, and pose health risks due to excessive chemical use.

Recent advancements in artificial intelligence and digital agriculture have introduced new opportunities to support farmers through intelligent crop monitoring systems. AI-based crop disease detection systems using computer vision and deep learning techniques can automatically analyze plant images and identify possible diseases or pest infestations. Convolutional neural networks (CNNs) have demonstrated strong performance in plant disease detection tasks by learning visual patterns from crop images [1], [2].

Despite these technological advancements, many existing agricultural diagnostic systems focus mainly on disease identification rather than providing actionable treatment guidance. Farmers who receive a disease diagnosis often still need to search for treatment recommendations from external agricultural resources. In addition, many systems rely only on image-based inputs and do not provide interactive advisory support for farmers who wish to describe symptoms using natural language or voice interaction.

To address these challenges, the PaddyGuard AI research project proposes a multimodal intelligent system for rice disease and pest management. The system integrates multiple technologies, including image-based disease detection, voice-based symptom analysis, and knowledge-driven treatment advisory mechanisms. Within this platform, the Treatment Advisory System and Agricultural Chatbot component play a critical role in assisting farmers after the diagnosis stage by providing context-aware treatment recommendations and preventive farming guidance.

The proposed system aims to bridge the gap between automated crop diagnosis and practical agricultural decision support. By combining artificial intelligence with domain knowledge from agricultural datasets such as the IRRI Rice Knowledge Bank and FAO crop protection guidelines, the system seeks to provide accurate and accessible treatment advice to farmers.

This research contributes to Sustainable Development Goal (SDG) 2: Zero Hunger by improving crop productivity and supporting sustainable food production systems. It also contributes to SDG 3: Good Health and Well-being by encouraging safe pesticide usage and reducing harmful chemical exposure. Furthermore, the research aligns with the Artificial Intelligence and Agricultural Informatics research domain, which focuses on developing intelligent technologies for smart agriculture and digital farming solutions.

1.2 Literature Review

Recent advancements in artificial intelligence (AI) and digital agriculture have significantly improved the development of intelligent systems that support farmers in crop monitoring and decision-making. Agricultural Decision Support Systems (DSS) have been widely studied as tools that integrate data analysis, expert knowledge, and computational models to assist farmers in diagnosing crop problems and selecting appropriate treatment strategies. Deep learning techniques, particularly convolutional neural networks (CNNs), have shown strong performance in identifying plant diseases from leaf images and other visual crop data [1], [2]. These technologies enable automated analysis of agricultural data and help farmers detect crop problems at early stages.

Several studies have explored knowledge-based agricultural advisory systems designed to provide crop management guidance to farmers. Such systems typically rely on predefined agronomic rules and expert knowledge to generate treatment recommendations. Rule-based agricultural advisory platforms demonstrate the potential of knowledge-driven decision support in assisting farmers with crop management decisions. However, these systems often struggle to handle diverse user queries and complex agricultural conditions due to their limited flexibility and dependence on manually defined rules. As agricultural environments involve numerous variables such as crop stage, weather conditions, and pest severity, purely rule-based approaches may not always provide adaptive and context-aware recommendations.

In recent years, digital agricultural advisory platforms have emerged to improve farmers' access to agricultural knowledge and crop protection information. Applications such as Plantix and Agrio provide farmers with disease detection capabilities and basic treatment recommendations through mobile and web platforms. These systems typically combine image-based crop disease identification with general advisory information. While such platforms have improved accessibility to agricultural knowledge, many systems still provide generalized treatment suggestions without considering contextual factors such as environmental conditions, crop growth stages, or pesticide dosage recommendations. This limitation reduces the effectiveness of advisory systems in real farming scenarios.

Another important research direction involves the use of conversational artificial intelligence for agricultural advisory services. Chatbot-based agricultural systems allow farmers to interact with digital platforms using natural language queries and receive automated responses to farming-related questions. Natural language processing (NLP) technologies enable chatbots to interpret user questions and retrieve relevant agricultural information. Recent advancements in transformer-based language models have significantly improved the ability of conversational systems to understand complex queries and generate accurate responses. Models such as BERT have been widely adopted to enhance language understanding in conversational systems and information retrieval applications [4].

Furthermore, retrieval-based knowledge systems have been introduced to improve the reliability of chatbot responses in domain-specific applications. Retrieval-Augmented Generation (RAG) combines neural language models with external knowledge retrieval mechanisms to generate responses based on trusted information sources rather than relying solely on model training data [5]. This approach enables advisory systems to access structured agricultural knowledge bases and provide more accurate and explainable responses for farmers.

Despite these advancements, several limitations remain in existing agricultural advisory systems. Many current solutions focus primarily on crop disease detection using computer vision models while providing limited support for treatment recommendations and decisionmaking guidance. In addition, many advisory platforms rely on static knowledge bases and lack intelligent mechanisms to integrate diagnostic results with context-aware treatment advice. These limitations highlight the need for more integrated and farmer-centric digital advisory systems that combine AI-based diagnosis, knowledge-driven recommendation mechanisms, and conversational interfaces to support effective crop management.

1.3 Research Gap

Although artificial intelligence has been widely applied in crop disease detection and digital agricultural advisory services, several important limitations remain in existing systems. Many previous studies have primarily focused on automated disease identification using image-based deep learning models. Convolutional neural networks and other computer vision techniques have demonstrated strong performance in identifying plant diseases from crop images [1], [2]. However, these systems often concentrate mainly on disease classification and do not provide detailed treatment recommendations that can effectively support farmers in real-world crop management.

Another important limitation observed in existing research is the lack of integration between crop diagnosis systems and agricultural advisory mechanisms. In many current platforms, disease detection and treatment guidance are implemented as separate processes. As a result, farmers are required to manually interpret diagnostic results and search for treatment information from different agricultural resources. This fragmented workflow can lead to incorrect pesticide usage, delayed treatment decisions, and reduced crop productivity.

Furthermore, many existing agricultural advisory platforms provide generalized recommendations that do not consider important contextual factors such as crop growth stage, environmental conditions, or disease severity. Without incorporating such contextual information, treatment recommendations may not always be appropriate for the specific farming situation. In addition, several existing systems lack interactive conversational interfaces that allow farmers to describe crop symptoms or ask questions using natural language.

Recent research in natural language processing and knowledge-based systems has introduced approaches such as Retrieval-Augmented Generation (RAG), which combines language models with external knowledge retrieval to improve response accuracy in domain-specific applications [5]. However, the application of such knowledge-driven conversational systems in agricultural advisory platforms remains limited.

Therefore, there is a clear need for a more integrated and intelligent advisory system that bridges the gap between automated crop diagnosis and practical treatment guidance. The proposed **Treatment Advisory System and Agricultural Chatbot** within the PaddyGuard AI platform aim to address these limitations by developing a web-based intelligent decision support system that integrates AI-generated diagnostic results with a knowledge-driven treatment advisory engine and conversational chatbot interface. By combining natural language processing techniques, knowledge retrieval mechanisms, and context-aware recommendation models, the proposed system aims to provide more accurate, accessible, and farmer-friendly agricultural guidance for rice disease and pest management.

This research aims to bridge the gap between automated crop diagnosis and practical agricultural decision support by developing a knowledge-driven Treatment Advisory System integrated with a conversational Agricultural Chatbot. The proposed system utilizes natural language processing, retrieval-based knowledge mechanisms, and rule-based validation to provide farmers with reliable treatment recommendations following crop diagnosis. By combining AI-based diagnostic outputs with structured agricultural knowledge sources such as the IRRI Rice Knowledge Bank and FAO guidelines, the system enables farmers to obtain accurate, context-aware crop management guidance through an accessible conversational interface.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop a web-based intelligent treatment advisory system and agricultural chatbot that provides farmers with accurate, context-aware recommendations for managing rice diseases and pest infestations. The system aims to integrate a knowledge-based agricultural dataset, natural language processing techniques, and retrieval-based advisory mechanisms within the PaddyGuard AI decision support platform to assist farmers in selecting appropriate treatment methods and improving crop management decisions.

2.2 Specific Objectives

- To collect and organize a structured agricultural knowledge dataset containing rice disease and pest treatment guidelines, pesticide recommendations, and preventive farming practices from reliable sources such as the IRRI Rice Knowledge Bank, FAO crop protection guidelines, and PlantVillage disease metadata.

- To design and implement a knowledge-based treatment advisory module that generates appropriate treatment recommendations based on disease or pest diagnosis results.

- To develop an agricultural chatbot using natural language processing techniques that allows farmers to interact with the system through natural language queries and receive relevant treatment guidance.

- To integrate a retrieval-based knowledge mechanism that enables the chatbot to access agricultural knowledge and provide accurate and context-aware responses.

- To evaluate the effectiveness of the proposed advisory system using performance indicators such as response accuracy, recommendation relevance, and usability of the chatbot interface.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science and Experimental research approach to design, develop, and evaluate a Treatment Advisory System and Agricultural Chatbot within the PaddyGuard AI platform. Design Science Research (DSR) is appropriate for this study because the primary goal is to develop a technological artifact that addresses a real-world agricultural problem—specifically, the lack of accessible and reliable treatment guidance for rice diseases and pest infestations.

The artifact developed in this research is a web-based advisory system integrated with a conversational agricultural chatbot that provides treatment recommendations for rice diseases and pest management. The system aims to assist farmers in interpreting diagnostic results and obtaining accurate guidance for crop protection.

The experimental component of the research focuses on evaluating the effectiveness of the advisory system in generating accurate treatment recommendations. The research process involves several stages, including dataset preparation, knowledge base construction, chatbot implementation, system integration, and performance evaluation. The developed advisory module will then be integrated with other components of the PaddyGuard AI system to provide comprehensive decision support for rice disease and pest management.

3.2 Technical Methods and Algorithms

The proposed advisory system integrates several computational techniques, including Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), semantic search, and rule-based advisory mechanisms to generate accurate treatment recommendations.

The agricultural chatbot serves as the main user interaction interface. Farmers can ask questions related to crop symptoms, pest infestations, or treatment methods using natural language queries. The system processes these queries using NLP techniques such as text normalization, tokenization, and keyword extraction in order to identify relevant agricultural concepts.

To improve the accuracy of responses, the system implements a Retrieval-Augmented Generation (RAG) mechanism. In this approach, the chatbot retrieves relevant information from a curated agricultural knowledge base before generating responses. This ensures that chatbot responses are grounded in verified agricultural knowledge sources rather than relying only on generative models.

The knowledge base contains structured agricultural information such as disease symptoms, pest descriptions, treatment guidelines, pesticide recommendations, and preventive farming practices. A semantic search mechanism based on vector similarity will be used to retrieve the most relevant documents when a farmer submits a query.

In addition, a rule-based advisory module will be implemented to ensure that treatment recommendations follow established agricultural guidelines. This module maps diagnosis results (disease or pest types) to recommended treatment strategies, including pesticide usage, biological control methods, and preventive practices.

The integration of conversational AI, knowledge retrieval techniques, and rule-based validation enables the system to provide accurate and context-aware treatment recommendations.

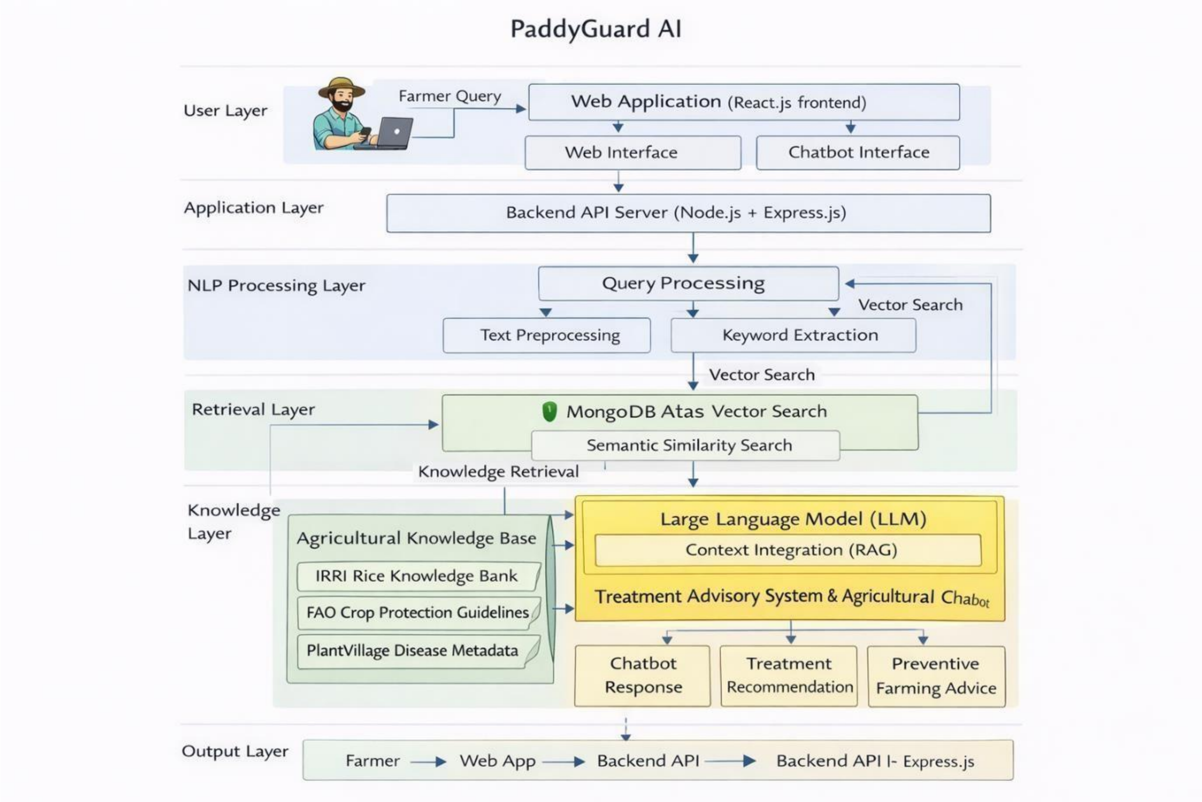

Figure 1: RAG Architecture of the PaddyGuard AI Advisory Chatbot

3.3 Tools and Platforms

The proposed advisory system will be implemented using modern web development and artificial intelligence technologies.

The frontend of the system will be developed using React.js, which allows the creation of responsive and interactive user interfaces that can be accessed through mobile devices and web browsers. This interface will enable farmers to interact with the chatbot and view treatment recommendations.

The backend API will be developed using Node.js and Express.js, which provide a lightweight and scalable framework for building RESTful web services. The backend server will manage user queries, handle communication between the chatbot module and the knowledge base, and return advisory responses to the frontend application.

Natural Language Processing and conversational AI functionalities will be implemented using frameworks such as LangChain or Hugging Face Transformers, which support the development of chatbot systems and retrieval-based knowledge models.

For storing agricultural knowledge and advisory rules, a MongoDB database will be used. MongoDB allows flexible storage of agricultural documents, including disease descriptions, pest management strategies, and treatment guidelines.

Development and experimentation tasks may also be conducted using Jupyter Notebook or Google Colab, which provide convenient environments for testing natural language processing workflows and evaluating system performance.

3.4 Data Collection and Ethical Considerations

The advisory system will utilize a curated agricultural knowledge dataset compiled from reliable agricultural information sources. These datasets provide verified information related to rice diseases, pest management practices, pesticide usage, and crop protection strategies.

The primary dataset used in this research is the Rice Knowledge Bank developed by the International Rice Research Institute (IRRI), which contains comprehensive information about rice crop diseases, pests, symptoms, and treatment recommendations.

In addition to IRRI data, the Food and Agriculture Organization (FAO) crop protection guidelines will be used to obtain standardized pesticide usage recommendations and crop protection practices. These guidelines help ensure that treatment recommendations generated by the advisory system follow internationally recognized agricultural standards.

Furthermore, PlantVillage disease metadata will be used as a supplementary reference dataset to provide additional information about disease types and crop conditions. Although PlantVillage is mainly used for plant disease detection research, its disease metadata provides useful contextual information for linking diagnosis results with treatment recommendations.

All collected information will be organized into a structured knowledge base and indexed to enable efficient retrieval during chatbot interactions.

From an ethical perspective, the research relies exclusively on publicly available agricultural datasets and institutional knowledge repositories. No personal or sensitive data will be collected during the research process. If user queries are recorded for system evaluation purposes, they will be anonymized to ensure user privacy and confidentiality.

3.5 Validation Strategy and Performance Metrics

The performance of the proposed treatment advisory system and chatbot will be evaluated using a structured validation strategy. The evaluation focuses on measuring the accuracy, relevance, and usability of the system's responses.

The validation process will involve testing the chatbot with a set of predefined agricultural queries related to rice diseases, pest management, pesticide usage, and crop protection practices. The responses generated by the chatbot will be compared with recommendations provided by established agricultural guidelines to determine the correctness of the advisory information.

Several performance metrics will be used to evaluate system effectiveness. Response accuracy will measure whether the chatbot provides correct treatment recommendations based on the farmer's query. Recommendation relevance will evaluate how well the generated responses address the specific agricultural problem described by the user.

Additionally, response time will be measured to assess the efficiency of the system in retrieving knowledge and generating responses. A lower response time indicates better system performance and improved user experience.

To further evaluate system usability, qualitative analysis may also be conducted by reviewing chatbot responses and comparing them with expert agricultural recommendations. This evaluation will help determine whether the advisory system provides practical and reliable guidance for farmers managing rice diseases and pest infestations.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed System

The PaddyGuard AI system is designed as a multimodal intelligent decision support platform for rice disease and pest management. The system integrates several artificial intelligence components that assist farmers in diagnosing crop problems and obtaining appropriate treatment guidance. The major components of the system include image-based disease detection, pest identification, voice-based symptom analysis, and a treatment advisory system supported by an agricultural chatbot.

The Treatment Advisory System and Agricultural Chatbot component play a critical role after the crop diagnosis stage. Once a disease or pest is identified by the detection modules, the advisory component provides farmers with recommended treatment strategies, pesticide usage guidelines, and preventive farming practices.

The advisory system uses a knowledge-based approach combined with conversational AI to provide context-aware responses to farmers. The chatbot interface allows farmers to interact with the system using natural language queries and receive reliable treatment guidance based on verified agricultural knowledge sources such as the IRRI Rice Knowledge Bank and FAO crop protection guidelines.

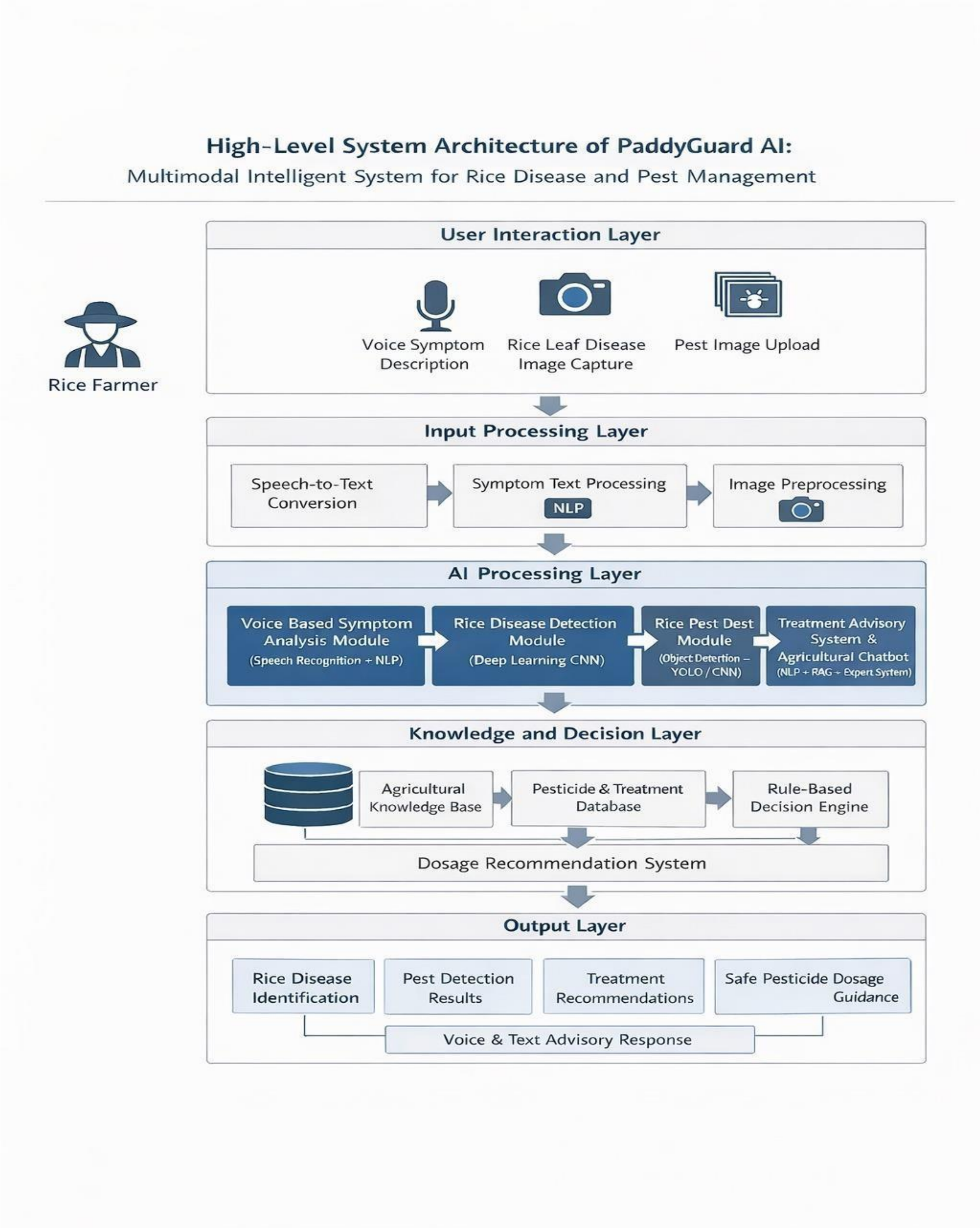

Figure 2:High-Level System Architecture of PaddyGuard AI

4.2 System Components

The overall PaddyGuard AI architecture consists of several interconnected components that work together to support intelligent crop management.

User Interface (Web Application)

The user interface allows farmers to interact with the system through a web-based platform. Farmers can upload crop images, describe symptoms, or ask questions related to crop diseases and pest management.

Diagnosis Modules

These modules include image-based disease detection models and pest identification systems developed using deep learning techniques. The diagnosis modules analyze crop images and provide predicted disease or pest results.

Treatment Advisory System

The treatment advisory module generates appropriate management recommendations based on the diagnostic results. The module uses agricultural guidelines and rule-based knowledge to suggest treatment strategies, pesticide usage recommendations, and preventive farming practices.

Agricultural Chatbot

The chatbot provides an interactive conversational interface that allows farmers to ask questions about crop diseases, pests, and treatment methods. Natural language processing techniques are used to understand user queries and retrieve relevant agricultural knowledge.

Knowledge Base

The knowledge base stores structured agricultural information collected from sources such as the IRRI Rice Knowledge Bank, FAO guidelines, and PlantVillage metadata. This information includes disease symptoms, pest descriptions, treatment recommendations, and crop management practices.

Backend API Server

The backend server manages communication between system components. It processes user requests, retrieves knowledge from the database, and sends advisory responses to the user interface.

4.3 Treatment Advisory System and Chatbot Component

The Treatment Advisory System and Agricultural Chatbot component consists of several internal modules that work together to generate treatment recommendations.

Query Processing Module

This module processes user queries submitted through the chatbot interface. Natural language processing techniques are applied to identify keywords related to diseases, pests, or treatment requests.

Knowledge Retrieval Module

The system retrieves relevant agricultural information from the knowledge base using semantic search techniques. Retrieval-Augmented Generation (RAG) mechanisms are used to combine knowledge retrieval with response generation.

Advisory Rule Engine

The advisory rule engine validates treatment recommendations using predefined agricultural rules and guidelines. This ensures that recommendations follow reliable crop protection practices.

Response Generation Module

This module generates the final advisory response that is presented to the farmer through the chatbot interface.

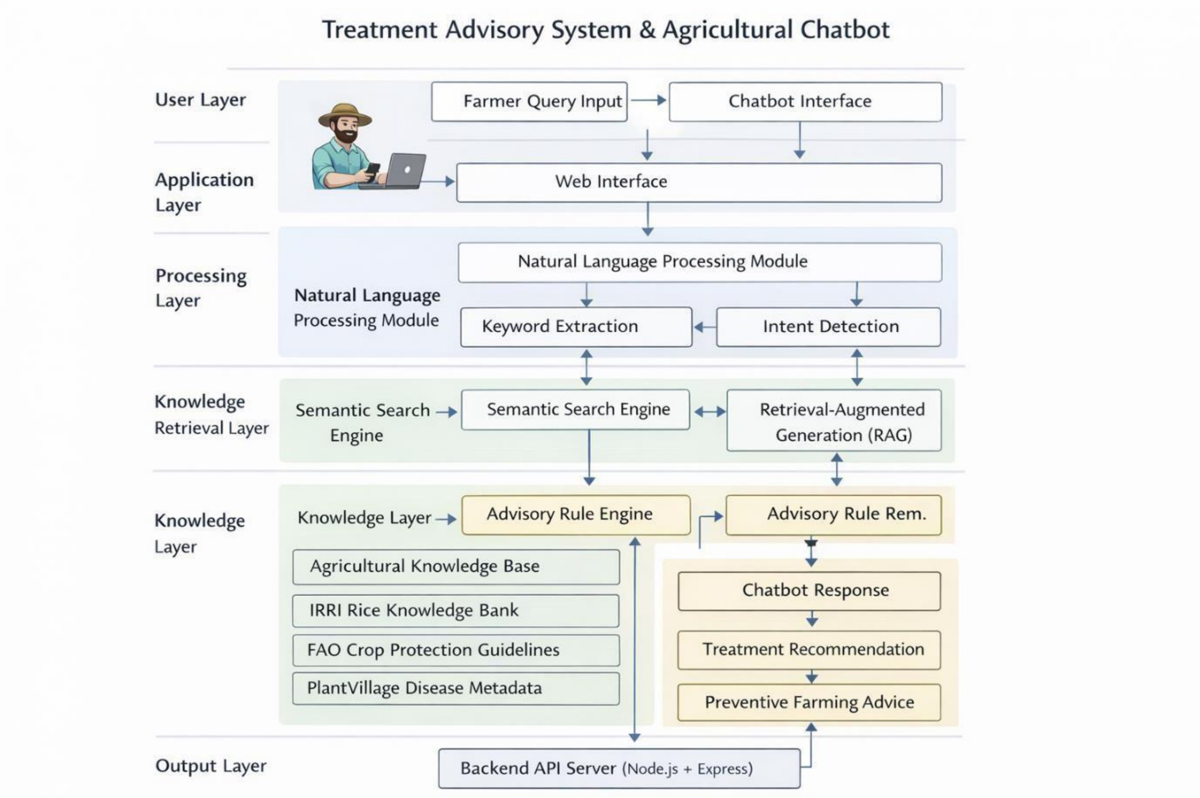

Figure 3: Architecture of the Treatment Advisory System and Agricultural Chatbot

4.4 Component Interaction and Data Flow

The data flow within the PaddyGuard AI system begins when a farmer interacts with the web application. The user may either upload an image of a crop symptom or submit a question related to crop health through the chatbot interface.

If an image is uploaded, the system forwards the image to the disease detection or pest identification module for analysis. Once the system identifies the possible disease or pest, the result is sent to the treatment advisory module.

If a user submits a question directly through the chatbot interface, the query is processed using natural language processing techniques. The system extracts relevant keywords and retrieves related agricultural knowledge from the knowledge base.

The retrieved information is then passed to the advisory rule engine, which generates appropriate treatment recommendations. The final response is delivered to the farmer through the chatbot interface in a clear and user-friendly format.

This architecture enables seamless integration between crop diagnosis modules and the advisory system, ensuring that farmers receive both diagnosis results and treatment guidance in a single platform.

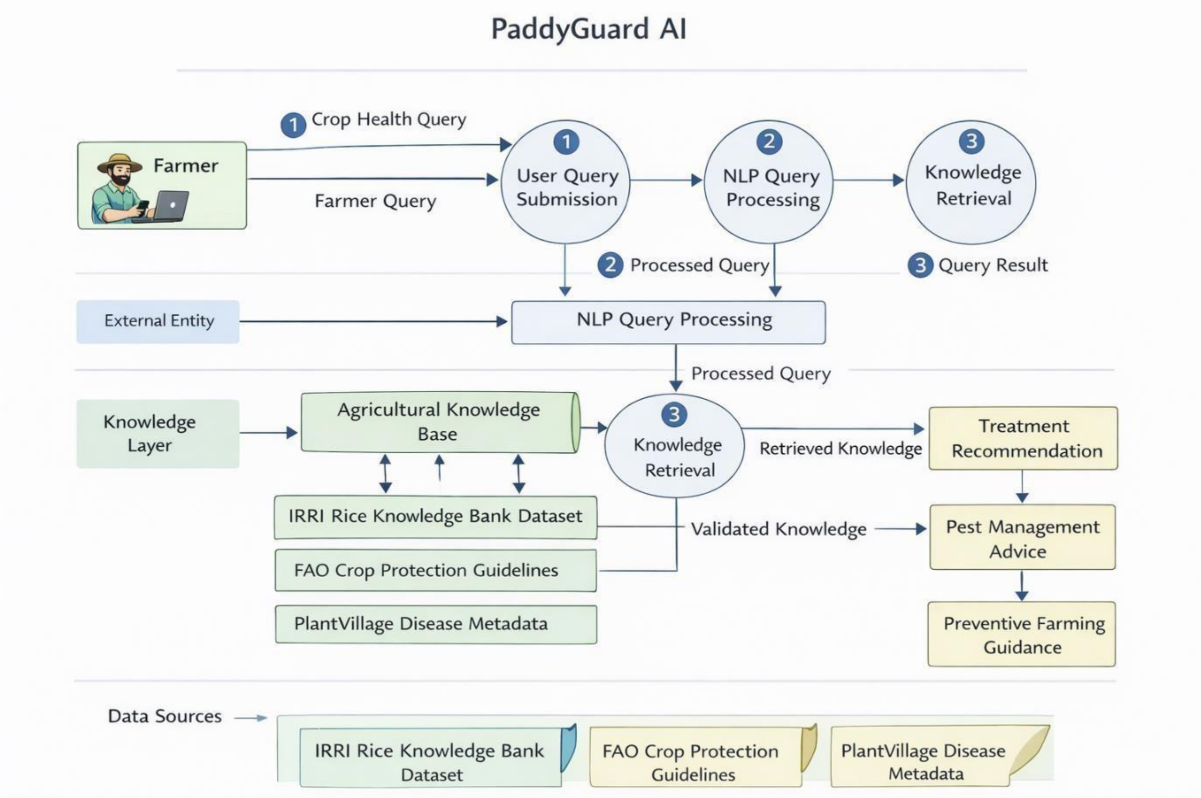

Figure 4: Data Flow Diagram of the PaddyGuard AI Advisory System

4.5 Scalability and Security Considerations

The proposed system is designed to be scalable and capable of supporting multiple users simultaneously. The use of web-based architecture and modular components allows the system to be expanded as new features or datasets become available. For example, additional crop types or disease categories can be integrated into the knowledge base without significantly modifying the system architecture.

Cloud-based deployment can further improve system scalability by allowing computational resources to be dynamically allocated based on system demand. This ensures that the system remains responsive even when multiple farmers access the platform simultaneously.

Security considerations are also important in the design of the system. User interactions with the web platform will be secured using standard web security practices such as secure API communication and data encryption. Although the system primarily uses publicly available agricultural knowledge datasets, any user-generated data or interaction logs will be handled with appropriate privacy protections.

By incorporating scalable architecture and secure system design, the PaddyGuard AI platform aims to provide a reliable and sustainable digital advisory solution for farmers.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The PaddyGuard AI system is designed to support multiple stakeholders involved in rice cultivation, agricultural advisory services, and crop monitoring activities. These stakeholders can be categorized into three main groups based on their level of interaction with the system.

- **Primary Stakeholders:**

The primary stakeholders are rice farmers, particularly smallholder farmers who rely on paddy cultivation as their primary source of livelihood. Many farmers face difficulties in accurately identifying pest infestations due to limited access to agricultural experts and diagnostic tools. The proposed system provides these farmers with a simple and accessible tool to upload pest images and receive diagnostic feedback, enabling them to take timely actions to protect their crops.

- **Secondary Stakeholders:**

Agricultural extension officers, crop advisors, and agronomists represent the secondary stakeholder group. These professionals can use the system as a supporting decision-making tool when assisting farmers in identifying pest infestations and recommending appropriate crop management strategies. The system can also help extension officers provide faster and more consistent advisory services during field visits.

- **Tertiary Stakeholders:**

Government agricultural institutions such as the Department of Agriculture, agricultural research organizations, and agricultural technology developers form the tertiary stakeholder group. These stakeholders may benefit from aggregated system data to analyze pest occurrence patterns, monitor crop health trends, and improve future agricultural technologies and policy decisions related to crop protection and food security.

5.2 Functional Requirements

The functional requirements describe the key features and operations that the PaddyGuard AI advisory system must support.

FR1: User Query Submission

The system shall allow farmers to submit questions related to rice diseases, pest infestations, and treatment practices through the chatbot interface.

FR2: Image Upload for Diagnosis

The system shall allow users to upload crop images for disease or pest detection.

FR3: Natural Language Query Processing

The system shall process farmer queries using natural language processing techniques to understand the user's request.

FR4: Knowledge Retrieval

The system shall retrieve relevant agricultural information from the knowledge base using semantic search and retrieval mechanisms.

FR5: Treatment Recommendation Generation

The system shall generate treatment recommendations based on diagnosis results and agricultural knowledge guidelines.

FR6: Advisory Rule Validation

The system shall validate treatment recommendations using rule-based agricultural guidelines to ensure safe and accurate advice.

FR7: Chatbot Response Generation

The system shall provide responses to user queries through a conversational chatbot interface.

FR8: Preventive Farming Advice

The system shall provide preventive recommendations to reduce the risk of future pest infestations and crop diseases.

FR9: Knowledge Base Management

The system shall allow administrators to update and manage the agricultural knowledge base used by the advisory system.

5.3 Non-Functional Requirements

Non-functional requirements describe the quality attributes and performance expectations of the PaddyGuard AI system.

Security

The system shall ensure secure communication between the frontend and backend services using secure API protocols. Any user data collected by the system shall be protected through appropriate data privacy measures.

Performance

The system shall provide chatbot responses within a short response time to ensure efficient user interaction. The advisory system should handle multiple user requests simultaneously without significant performance degradation.

Usability

The user interface shall be simple and easy to use for farmers with limited technical knowledge. The chatbot interface should support natural language interaction and present recommendations in a clear and understandable format.

Scalability

The system architecture shall support future expansion to include additional crop types, diseases, or advisory modules without major modifications to the existing system.

Reliability

The system shall provide consistent and reliable advisory responses by retrieving information from verified agricultural knowledge sources such as IRRI and FAO guidelines.

Maintainability

The system shall be designed using modular architecture so that system components can be updated or improved without affecting the overall platform functionality.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the PaddyGuard AI platform consists of rice farmers and agricultural stakeholders involved in crop management and advisory services. In many riceproducing regions, farmers face challenges in accessing timely agricultural expertise and reliable treatment guidance for crop diseases and pest infestations.

The system primarily targets smallholder rice farmers, who often have limited access to agricultural experts or laboratory testing facilities. These farmers require simple and accessible digital tools that help them identify crop problems and obtain clear treatment recommendations.

In addition to farmers, agricultural extension officers and agribusiness organizations represent secondary customer segments. Extension officers can use the platform as a digital decisionsupport tool to assist farmers in diagnosing crop problems and recommending appropriate pest management strategies.

A typical user persona is a small-scale rice farmer who owns a smartphone and seeks quick guidance for identifying crop diseases and selecting safe treatment methods. The farmer interacts with the PaddyGuard AI system through a web or mobile interface to upload crop images or ask questions through the chatbot.

6.2 Value Proposition

The PaddyGuard AI system provides significant value by combining artificial intelligencebased diagnosis with knowledge-driven treatment advisory services in a single platform.

Unlike many existing agricultural applications that focus only on disease detection, PaddyGuard AI integrates diagnosis results with context-aware treatment recommendations. The treatment advisory system and agricultural chatbot provide farmers with detailed guidance on pesticide usage, disease management strategies, and preventive farming practices.

The platform also improves accessibility through a conversational chatbot interface, enabling farmers to ask questions in natural language and receive clear and understandable responses. This approach reduces the need for farmers to manually search for treatment information from multiple sources.

By providing reliable and explainable recommendations, the system helps farmers make better crop management decisions, reduce crop losses, and minimize unsafe pesticide usage.

6.3 Revenue Model

The PaddyGuard AI platform can adopt multiple revenue models to ensure financial sustainability and scalability.

One possible model is a freemium approach, where basic advisory services are provided free of charge to farmers, while advanced features such as detailed crop analytics, personalized recommendations, or premium advisory services are offered through a subscription plan.

Another potential revenue stream is institutional partnerships with agricultural organizations, government agencies, or agricultural technology companies that may integrate the advisory platform into their digital farming initiatives.

Additionally, the system could generate revenue through data-driven agricultural insights, where aggregated crop health data can help agribusiness companies understand disease trends and improve agricultural supply chains.

6.4 Cost Estimation

The implementation and deployment of the PaddyGuard AI advisory system involve several cost components.

Development costs include software development, system integration, and AI model implementation. These costs primarily involve the time and effort required by software engineers and researchers working on the project.

Infrastructure costs include cloud computing resources required for hosting the web application, running AI models, and storing agricultural knowledge datasets. Cloud services may be required to ensure system scalability and availability.

Maintenance costs include system updates, knowledge base improvements, security updates, and technical support required to ensure long-term system reliability.

6.5 Pricing Strategy

The pricing strategy for the PaddyGuard AI platform should consider the financial capabilities of smallholder farmers while ensuring system sustainability.

A tiered pricing model may be adopted where basic advisory services are offered free or at a minimal cost, allowing farmers to access essential disease diagnosis and treatment recommendations.

Advanced features such as premium advisory services, advanced crop analytics, or personalized consultation support may be offered through subscription-based pricing.

Institutional users such as agricultural companies or government organizations may access enterprise-level services through customized pricing agreements.

6.6 Competitive Advantage

The PaddyGuard AI platform offers several advantages compared to existing agricultural diagnostic systems.

First, the system integrates multimodal inputs including image-based detection, voice-based symptom descriptions, and conversational chatbot interaction, providing a more flexible user experience.

Second, the platform combines AI-based predictions with knowledge-driven treatment advisory mechanisms, ensuring that farmers receive actionable guidance rather than only diagnostic results.

Third, the use of Retrieval-Augmented Generation (RAG) and expert rule validation improves the reliability and explainability of treatment recommendations.

These features collectively provide a more comprehensive and farmer-friendly agricultural decision support system.

6.7 Intellectual Property Considerations

The PaddyGuard AI platform involves several intellectual property components related to system design, algorithms, and data processing techniques.

The software architecture, chatbot advisory mechanisms, and rule-based treatment recommendation system may be protected through software copyrights and intellectual property rights.

In addition, proprietary datasets, knowledge base structures, and AI model configurations developed during the research process may represent valuable intellectual assets.

Any use of external datasets such as the IRRI Rice Knowledge Bank, FAO agricultural guidelines, or PlantVillage metadata will follow appropriate licensing and attribution requirements to ensure compliance with data usage policies.

Protecting intellectual property ensures that the developed technology can be further commercialized while maintaining compliance with legal and ethical standards.

7. BUDGET AND JUSTIFICATION

7.1 Estimated Project Budget

The development and deployment of the PaddyGuard AI treatment advisory system require several resources including computing hardware, software tools, cloud infrastructure, and human effort. The estimated budget allocation for the project is summarized below.

Category	**Estimated Cost (LKR)**
Hardware Costs	15,000
Software Licenses	5,000
Cloud Services	15,000
Human Effort Cost	3,000
Total Estimated Budget	**38,000**

Table 1: Estimated Project Budget

7.2 Hardware Costs

Hardware resources are required for developing, testing, and running the PaddyGuard AI system. Development machines are necessary to support web application development, database management, and AI model experimentation.

Hardware Item	**Description**	**Estimated Cost (LKR)**
Development Laptop Upgrade	Memory or storage upgrades to support development and AI processing	10,000

External Storage Device	Used for storing datasets, backups, and experimental results	5,000

Table 2: Hardware Costs

Total Hardware Cost: LKR 15,000

7.3 Software Licenses

The advisory system will be developed using widely available open-source technologies such as React.js for the user interface, Node.js for backend services, MongoDB for data storage, and NLP libraries for implementing chatbot capabilities. These technologies eliminate the need for expensive commercial software.

However, minor expenses may occur when using design or productivity tools that assist in interface prototyping and development activities.x

Software	**Purpose**	**Estimated Cost (LKR)**
Interface Design Tool (Figma / similar	Interface prototyping and design tools	2,500
Development Utilities	Optional tools or extensions used to improve coding productivity	2,500

Table 3: Software Licenses

Total Software License Cost: LKR 5,000

7.4 Cloud Services

To ensure that the advisory chatbot can be accessed through the web platform, a basic cloud environment is required. Cloud services allow the application backend and database to be hosted online and support reliable communication between system components.

Cloud Resource	Purpose	Estimated Cost (LKR)
Cloud Hosting (AWS / Azure / GCP)	Deployment of the web application	10,000
Database Hosting (MongoDB Atlas)	Storage of agricultural knowledge datasets	5,000

Table 4: Cloud Services

Total Cloud Service Cost: LKR 15,000

7.5 Human Effort Cost

Since this research project is conducted as part of an undergraduate academic study, no direct labor costs are included. However, a small allocation is included for report preparation and submission requirements.

Role	Task	Estimated Cost (LKR)
Printing & Binding	Final research report and documentation submission	3,000

Table 5: Human Effort Cost

Total Human Effort Cost: LKR 3,000

7.6 Justification of Expenses

The proposed budget reflects the minimum resources required to design and implement the PaddyGuard AI Treatment Advisory System and Agricultural Chatbot component. The majority of the development technologies used in this project are open-source platforms, which significantly reduces the overall software cost.

Hardware upgrades are allocated to support application development, data processing, and system testing activities. Cloud services are required to deploy the backend system and store the agricultural knowledge base used by the advisory chatbot.

Since the system is developed as part of an undergraduate research project, no direct development salary costs are included. Only minimal expenses related to documentation and report submission are considered. Overall, the estimated budget is sufficient to support the successful development and deployment of the advisory system within the scope of the research project.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) outlines the major tasks involved in the development of the PaddyGuard AI system and distributes responsibilities among the project team members. The tasks are organized according to the overall twelve-month research timeline to ensure systematic progress throughout the project lifecycle.

Each team member is responsible for developing a specific technical component of the system while collaborating on shared activities such as dataset preparation, system integration, evaluation, and final documentation. This structured task allocation helps maintain balanced workload distribution and ensures that the development of different modules progresses in parallel.

	Task Description	**Responsible Member**	**Timeline**
T1	Literature review and research problem analysis	All members	Month 1–2
T2	Dataset collection and preprocessing	All members	Month 2–3
T3	Rice disease detection model development	Member 1	Month 3–7
T4	Rice pest detection model development	Member 2	Month 3–7
T5	Voice-based symptom analysis module developed	Member 3	Month 4–8
T6	Treatment advisory and knowledge system development	Member 4	Month 5–8
T7	Integration of system modules	All members	Month 8–9
T8	System testing and performance evaluation	All members	Month 9–10
T9	System refinement and optimization	All members	Month 10–11

T10	Final documentation and research report preparation	All members	Month 11–12

Table 6: WBS

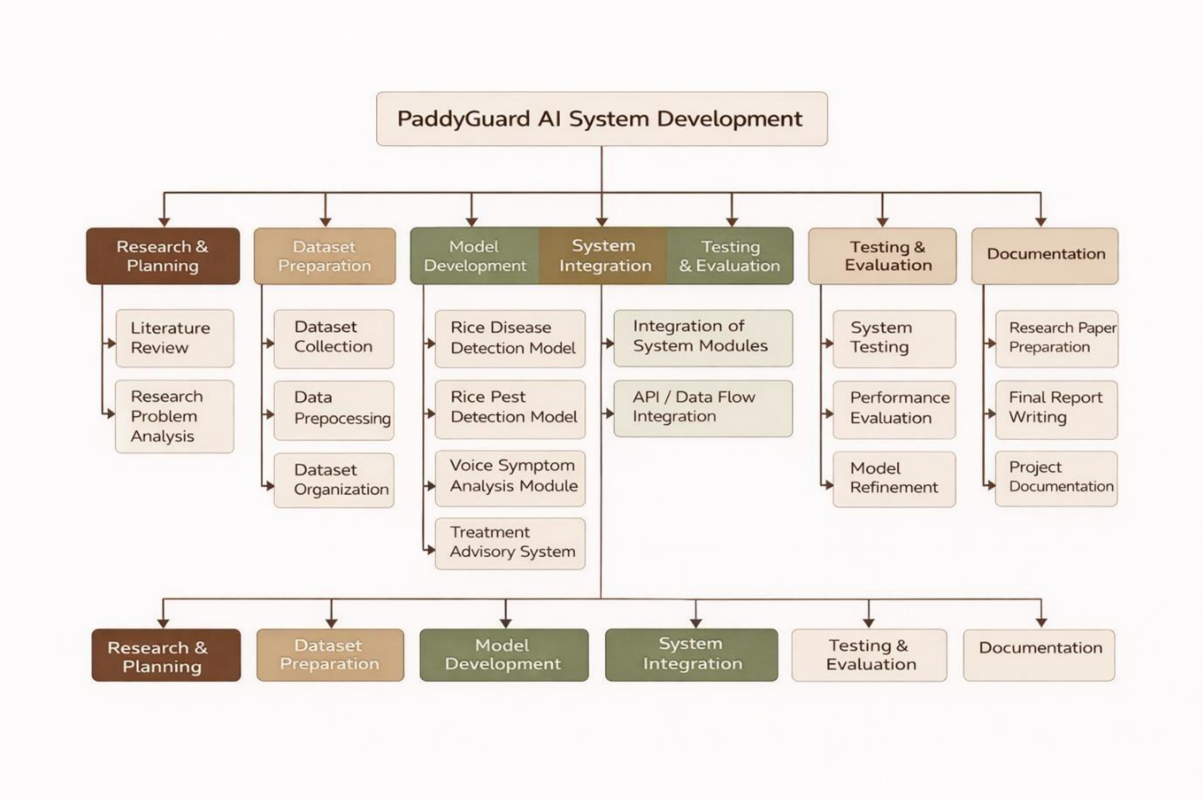

Figure 5:Work Breakdown Structure for PaddyGuard AI Development

The proposed work breakdown structure ensures that each member focuses on a clearly defined component while maintaining collaboration during the integration and evaluation phases of the project. This approach improves coordination among team members and supports the successful completion of the research objectives within the planned timeline.

9. GANTT CHART

The Gantt chart illustrates the timeline and scheduling of major tasks involved in the development of the PaddyGuard AI system. The project is planned over a twelve-month period, beginning with literature review and dataset preparation, followed by the development of individual system modules including disease detection, pest detection, voicebased symptom analysis, and the treatment advisory system.

Parallel development of the modules allows efficient progress while ensuring that each team member focuses on a specific technical component. The final stages of the project include system integration, testing, optimization, and the preparation of the final research report.

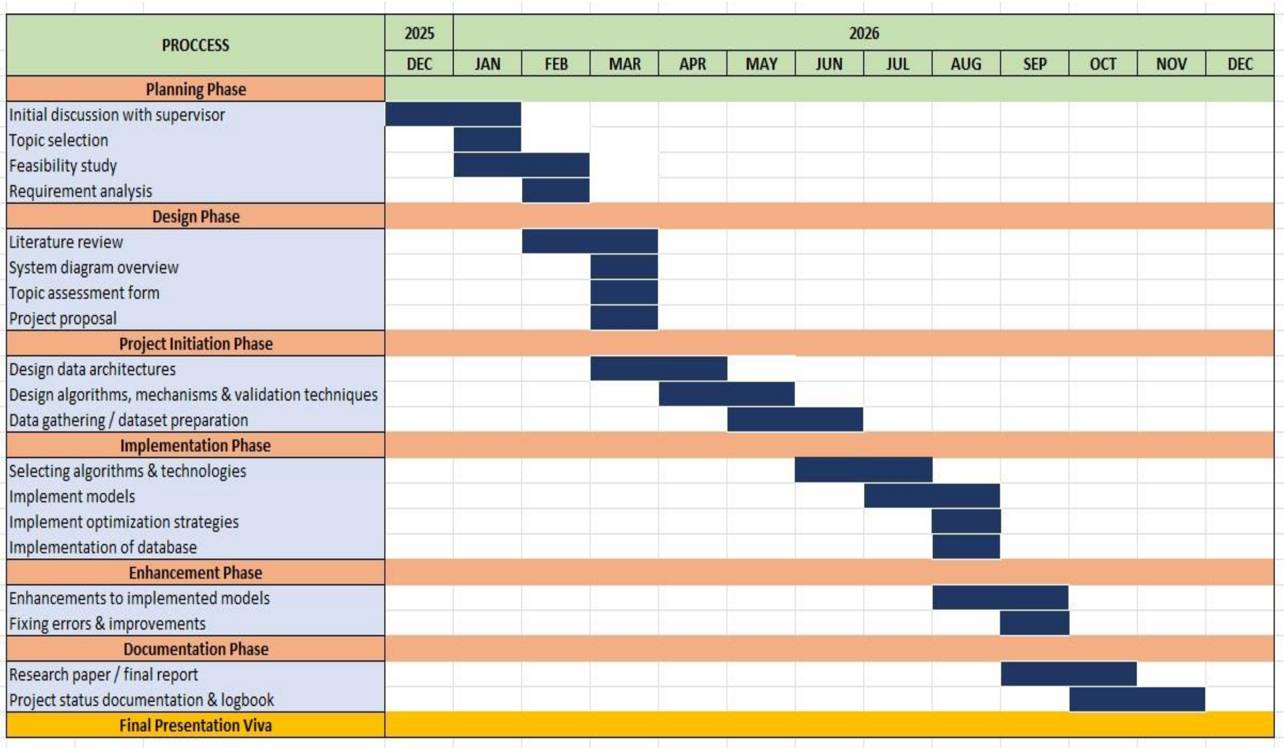

Figure 6: Gantt Chart

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APPENDICES

Appendix A – System Architecture Diagram

This appendix includes the high-level architecture diagram of the PaddyGuard AI platform. The diagram illustrates the interaction between major system components such as the disease detection module, pest detection module, voice-based symptom analysis module, and the treatment advisory system with the agricultural chatbot.

The architecture diagram also highlights the data flow between the user interface, backend API services, AI processing modules, and the agricultural knowledge base used for generating treatment recommendations.

Appendix B – Agricultural Knowledge Dataset Sources

This appendix provides details about the datasets and knowledge sources used for developing the treatment advisory system.

Dataset / Knowledge Source	**Description**
IRRI Rice Knowledge Bank	Provides authoritative information on rice diseases, pests, and treatment methods
FAO Crop Protection Guidelines	Contains international guidelines for safe pesticide usage and crop protection
PlantVillage Dataset	Public dataset containing plant disease images and agricultural metadata

These datasets support the knowledge-driven advisory system used by the chatbot to generate treatment recommendations.

Appendix C – Preliminary User Interface Mockups

Figure 7: Home Page

Figure 8: Leaf Diagnosis

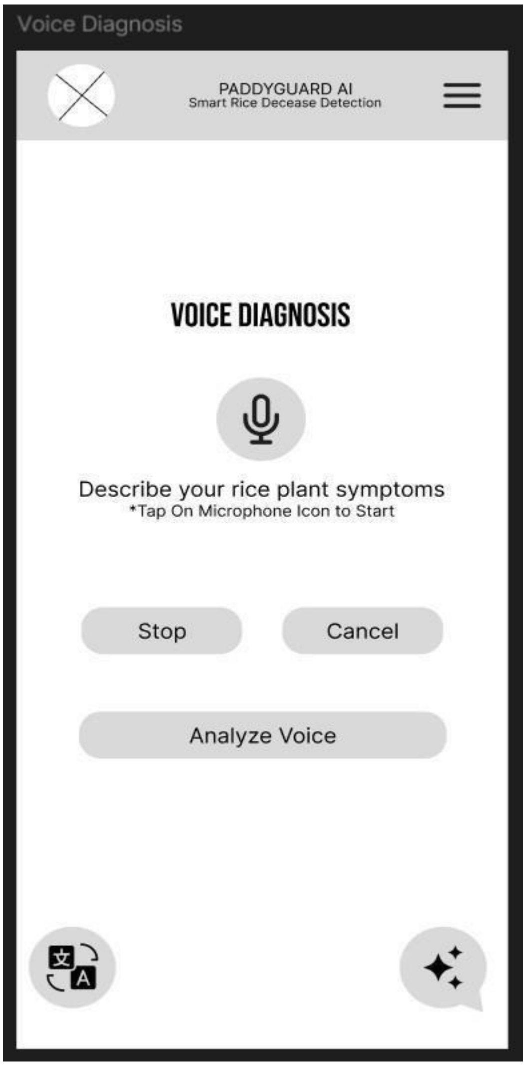

Figure 10: Voice Diagnosis

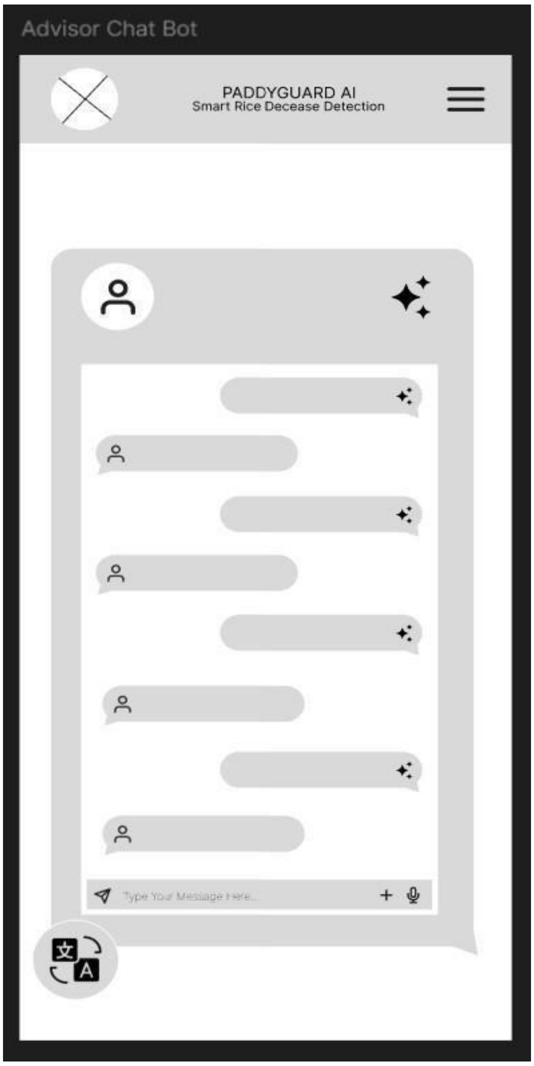

Figure 9: Advisory ChatBot

Figure 12: Pest Detection

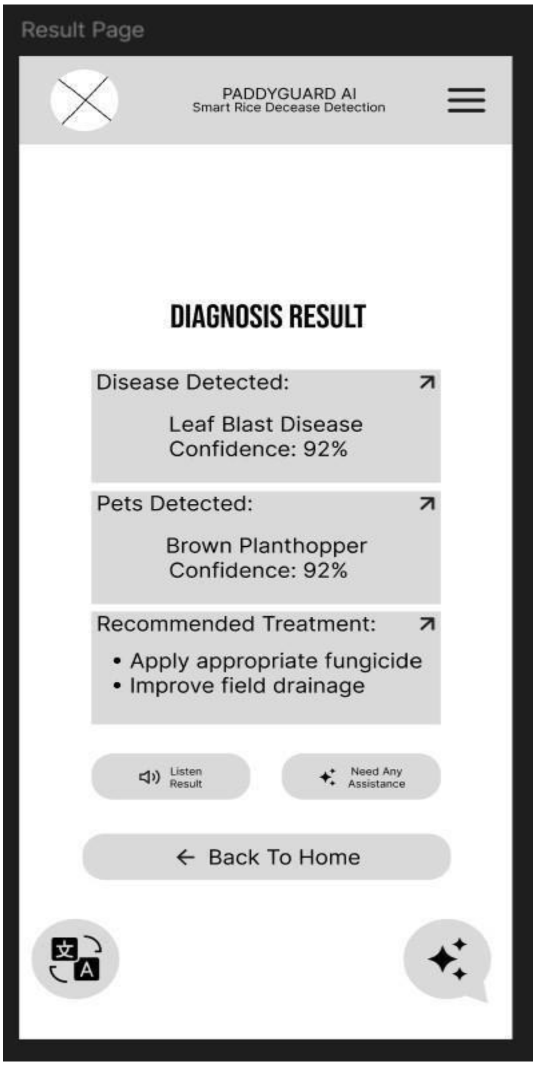

Figure 11: Result Page

Appendix D – Project Risk Analysis

This appendix presents the potential risks associated with the development and deployment of the PaddyGuard AI system.

Risk	**Description**	**Mitigation Strategy**
Data Quality Risk	Agricultural datasets may contain incomplete or inconsistent information	Use verified agricultural knowledge sources
Model Accuracy Risk	AI models may produce incorrect predictions	Use evaluation metrics and expert validation
System Integration Risk	Integration between system modules may cause technical issues	Conduct staged integration testing
User Adoption Risk	Farmers may be unfamiliar with digital advisory tools	Design simple and userfriendly interfaces

Appendix E – Ethical Considerations

This appendix outlines ethical considerations related to the research project.

The system uses publicly available agricultural datasets and does not collect personal or sensitive information from users. Any data used for research purposes will follow proper data usage policies and attribution requirements.

In addition, the system aims to promote responsible pesticide usage and sustainable farming practices by providing accurate and safe treatment recommendations based on trusted agricultural knowledge sources.